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ORIGINAL PAPER

Legal text summarization via judicial syllogism with large language models

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Abstract

Legal judgment documents are typically lengthy, complex, and filled with domain-specific language. Existing summarization methods often struggle to capture the essential legal reasoning, frequently omitting critical details such as the applicable law and the logical connection between facts and outcomes-resulting in summaries that lack interpretability and coherence. Recent advances in large language models (LLMs) have shown strong capabilities in multi-step reasoning and language generation, making them a promising foundation for tackling the challenges of legal summarization. However, their potential remains underutilized in the legal domain due to the need for explicit legal logic and structured inference. To address this issue, we propose a novel summarization framework based on judicial syllogism using large language models (LLMs). Specifically, we leverage the large language model guiding it explicitly to emulate classical judicial reasoning: extracting the major premise (applicable legal provisions), the minor premise (case-specific facts), and finally deducing the conclusion (judgment outcome). This structured reasoning is operationalized through carefully designed prompt engineering and fine-tuning techniques, enabling the model to reason step-by-step like a human judge. We evaluate our proposed method on two judicial summarization datasets. Experimental results demonstrate that our judicial syllogism approach achieves competitive performance compared to several strong baseline LLMs. Ablation studies further verify that each component (law, facts, conclusion) of the judicial syllogism contributes meaningfully to the summarization quality. Our findings clearly indicate that embedding explicit judicial syllogistic reasoning significantly enhances the accuracy and logical coherence of legal text summarization.

Keywords Natural language processing · Natural language generation · Legal text summarization · Judicial syllogism · Prompt engineering

1 Introduction

Summarizing legal documents such as court judgments is a crucial yet challenging task in legal AI (Akter et al. 2025). Judgment documents often span many pages and contain dense legal jargon and complex argumentation (Bauer et al. 2023). The judicial summary of a judgment is a condensed version that reflects the trial process, key facts, reasoning, and the final decision (Ragazzi et al. 2024). High-quality summaries help lawyers and the public quickly understand the essence of a case (Rush et al. 2015). However, producing such summaries automatically is difficult due to the need for domain expertise, logical reasoning, and fidelity to the source (Liu et al. 2024).

Legal language is highly domain-specific and presumes knowledge of laws and precedents (Chalkidis et al. 2021). Generic summarization models trained on news or general text often fail to capture crucial legal details and reasoning (Beltagy et al. 2020; See et al. 2017). Moreover, a good legal summary must not only extract facts but also explain why the court reached its decision - something that requires applying legal rules to facts, akin to a logical deduction (Nal-

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lapati et al. 2016). Traditional neural summarization methods (e.g., sequence-to-sequence models) tend to produce surface-level abridgements and may omit the legal rationale, resulting in summaries that lack the "because" behind the outcome (Nallapati et al. 2016; Elaraby et al. 2023). Ensuring logical consistency and completeness (covering facts and law) is thus a key challenge (Codrea 2024).

Despite these advances, existing methods still face three major limitations. First, they often treat summarization as a text-level compression task, ignoring the legal reasoning that underpins judicial decisions. Second, they lack explicit modeling of the logical structure - particularly the connection between legal rules and case facts - which leads to summaries that are factually correct but legally incomplete or ungrounded. Third, such summaries typically lack interpretability and fail to answer the core legal question: why did the court reach this outcome? These limitations highlight the need for a summarization framework that can explicitly model legal reasoning and generate summaries that reflect the deductive structure of judicial decisions.

In legal theory, judicial decision-making is often viewed as a syllogism - a form of deductive reasoning where a general rule (major premise) is applied to specific facts (minor premise) to derive a conclusion (Benhabib 1997). For example, a judge might use a law (major premise) and the established facts of the case (minor premise) to conclude the verdict. This classic logical structure is widely accepted in law and suggests a blueprint for summarization: a good summary should mention the governing law or principle, the relevant facts, and the outcome (Deng et al. 2024). We hypothesize that by explicitly modeling summarization after the judicial syllogism, we can guide an LLM to produce more informative and logically sound summaries (Wei et al. 2022). Notably, human judges are said to have their "gaze shuttling back and forth between facts and norms" during reasoning-our goal is to emulate this interplay in the summarization process (Yang et al. 2024).

In this paper, we introduce a Judicial Syllogism Summarization framework that integrates legal reasoning steps into the summarization process using a large language model (Xiao et al. 2021). We utilize Qwen, a state-of-the-art LLM, as the core summarization engine (Gao et al. 2021). We craft prompt-based chain-of-thought strategies that direct Qwen to first identify the applicable law (norm) and key facts, and then generate the summary conclusion (Wang et al. 2023). In effect, the model performs a step-by-step reasoning internally (or in a multi-stage pipeline) before producing the final summary (Bauer et al. 2023). This approach is implemented via prompt engineering (in-context instructions guiding the model's reasoning) and, in our extended setup, by fine-tuning Qwen on syllogism-structured data for further gains (Kojima et al. 2022; Yuan et al. 2021; Xiao et al. 2018; Dai et al. 2023). Our work makes the following contributions:

- We propose an innovative framework specifically tailored for legal text summarization, grounded in the classical judicial syllogism structure (law-fact-conclusion). To the best of our knowledge, this is the first study explicitly incorporating syllogistic reasoning into the summarization pipeline leveraging large language models.
- We design advanced prompt engineering techniques enabling judicial syllogism reasoning via chain-of-thought prompting. This method guides the large language model to explicitly articulate intermediate reasoning steps-identifying applicable laws and critical facts-prior to generating the final summary, thereby enhancing the logical coherence, interpretability, and accuracy of the generated summaries.
- We systematically evaluate our proposed approach on two benchmark datasets comprising authentic legal judgment documents. Extensive comparative experiments against several robust LLM baselines across multiple metrics demonstrate competitive performance of our judicial syllogism-based summarization framework.
- We conduct thorough ablation studies to empirically measure the contribution of each syllogistic reasoning component (law identification, fact extraction, and conclusion synthesis). Results confirm that explicitly modeling each reasoning step significantly improves the quality of summarization. Furthermore, detailed qualitative analyses and illustrative case studies highlight that our syllogism-guided approach generates summaries with clearer logical justification and greater interpretability compared to baselines.

The rest of this paper is organized as follows: Section 2 reviews related work in legal summarization and LLM reasoning. Section 3 details our methodology, including the syllogism-based framework, model design, and algorithms. Section 4 presents experiments, results on benchmarks, and ablation analyses. Finally, Section 5 concludes with insights and future directions for syllogism-inspired text summarization.

2 Related work

2.1 Legal document summarization

Summarization of legal texts has been explored via both extractive and abstractive methods (Yu et al. 2025; See et al. 2017; Nallapati et al. 2016). Early approaches often relied on rule-based extraction or statistical methods to identify important sentences (e.g., those containing case outcomes or legal terms) (Bauer et al. 2023). Recent neural approaches apply sequence-to-sequence models fine-tuned on legal data (Feijo and Moreira 2023). However, legal domain data for sum-

marization is scarce, and models trained on news struggle with legal jargon and structure (Liu et al. 2024; Xiao et al. 2018). To address domain gaps, researchers have created specialized datasets - for example, CAIL2020-SFZY provides thousands of Chinese court judgment-summary pairs (Ragazzi et al. 2024; Wang et al. 2023), and other efforts include EU cases or U.S. court opinion datasets (Heddaya et al. 2024). A notable trend is structure-aware summarization identified that Chinese court judgment summaries often follow a fixed structure (e.g., type of case, claims, arguments, reasoning, decision) and proposed to leverage this structure. Such methods explicitly segment the summary into sections like "Plaintiff's Claims" and "Court's Reasoning" to ensure key aspects are covered. Our work shares a similar spirit of enforcing coverage of crucial components, but we focus on the logical law-fact-conclusion syllogism rather than document sections (Wang et al. 2023; Martin 2011; Zhang and Song 2025; Hou et al. 2025).

2.2 Large language models in legal NLP

With the advent of large pre-trained language models, there has been growing interest in applying models like GPT-3, GPT-4, and their open counterparts to legal tasks (Achiam et al. 2023; Touvron et al. 2023). GPT-3.5/4 have demonstrated strong zero-shot or few-shot performance on tasks such as legal question answering and case outcome prediction, thanks to their vast knowledge and reasoning ability. However, for summarization, simply prompting a general LLM may yield generic outputs that miss legal specifics. Fine-tuning or specializing LLMs for law has shown promise: e.g., Legal-BERT variants and LawGPT models tuned on legal corpora (Chalkidis et al. 2020; Liu et al. 2024). More recently, several open-source LLMs focused on Chinese legal text have emerged: ChatGLM (GLM et al. 2024) (an instruction-tuned bilingual LM), Baichuan2 (Yang et al. 2023) and Qwen Yang et al. (2024) (by Alibaba, a powerful model with versions for Chinese and English). These models provide strong baselines and can be further adapted. In our experiments, we include GPT-4 as a representative proprietary model, and LLaMA2 and the mentioned Chinese LLMs as open alternatives, comparing their summarization performance to our approach.

2.3 Chain-of-thought reasoning

Chain-of-Thought Reasoning: Recent research in prompting LLMs has introduced chain-of-thought (CoT) prompting, which encourages the model to generate intermediate reasoning steps for complex tasks (Wei et al. 2022; Kojima et al. 2022; Zhou et al. 2022). CoT has significantly improved performance on arithmetic, commonsense, and multi-step reasoning tasks by allowing the model to break down problems into sub-tasks and solve them stepwise. In legal AI, multi-step reasoning aligns with how lawyers and judges deliberate: considering facts, applying laws, then concluding. For example, in legal judgment prediction tasks, multi-phase prompting frameworks have been proposed to emulate judge reasoning. Our work applies the chain-of-thought paradigm to summarization (Madaan et al. 2023; Deng et al. 2024; Yue et al. 2023). Instead of asking the model to output the summary in one go, we prompt it to first "think" about the case: identify the pertinent law (major premise) and facts (minor premise), and use those to form the final summary (conclusion). This approach is novel in summarization and bridges the gap between logical reasoning and text generation (Santosh et al. 2025; Martin 2011; Heddaya et al. 2024; Zheng et al. 2021; Almohaimeed and Azmi 2025; Sun et al. 2025; Zhang et al. 2025).

3 Methodology

In this section, we describe our judicial syllogism summarization framework in detail. We first formalize the task and then present our approach, which consists of multiple modules reflecting the major premise, minor premise, and conclusion steps. We also detail how we implement these modules using the Qwen LLM via chain-of-thought prompting. Pseudocode algorithms and mathematical formulations are provided for clarity.

3.1 Problem formulation

We frame legal text summarization as a conditional sequence generation task. Given a long-form judgment document $\mathbf{X} = (x_1, x_2, \ldots, x_n)$, the goal is to generate a concise yet faithful summary $\mathbf{Y} = (y_1, y_2, \ldots, y_m)$, where $m \ll n$. The generation model, typically a large language model (LLM) parameterized by θ, maximizes the conditional probability:

$$\hat{\mathbf{Y}} = \arg\max_{\mathbf{Y}} p_\theta(\mathbf{Y} \mid \mathbf{X}) = \prod_{t=1}^{m} p_\theta(y_t \mid \mathbf{X}, y_{<t}). \tag{1}$$

To train the model, we minimize the negative log-likelihood (equivalently, cross-entropy) between the generated and gold-standard summaries:

$$\mathcal{L}_{\text{CE}} = -\sum_{t=1}^{m} \log p_\theta(y_t^\star \mid \mathbf{X}, y_{<t}^\star). \tag{2}$$

Judicial syllogism as latent reasoning structure To better reflect legal reasoning, we introduce latent variables representing the syllogistic components: $\mathbf{P}_{\text{maj}}$ (major premise:

law) and $\mathbf{P}_{\min}$ (minor premise: facts). The summary generation is decomposed as:

$$p_\theta(\mathbf{Y}, \mathbf{Z} \mid \mathbf{X}) = p_\theta(\mathbf{P}_{\text{maj}} \mid \mathbf{X}) \cdot p_\theta(\mathbf{P}_{\min} \mid \mathbf{X}, \mathbf{P}_{\text{maj}}) \cdot p_\theta(\mathbf{Y} \mid \mathbf{X}, \mathbf{Z}), \tag{3}$$

where $\mathbf{Z} = \{\mathbf{P}_{\text{maj}}, \mathbf{P}_{\min}\}$. Since exact marginalization is intractable, we approximate using the most probable premises:

$$\tilde{\mathbf{Z}} = \arg\max_{\mathbf{Z}} p_\theta(\mathbf{Z} \mid \mathbf{X}), \quad \Rightarrow \quad p_\theta(\mathbf{Y} \mid \mathbf{X}) \approx p_\theta(\mathbf{Y} \mid \mathbf{X}, \tilde{\mathbf{Z}}). \tag{4}$$

Variational perspective Alternatively, we adopt a variational approach and optimize the evidence lower bound (ELBO) with a variational posterior $q(\mathbf{Z} \mid \mathbf{X})$:

$$\log p_\theta(\mathbf{Y} \mid \mathbf{X}) \geq \mathbb{E}_q \left[\log p_\theta(\mathbf{Y}, \mathbf{Z} \mid \mathbf{X})\right] - \text{KL}(q \parallel p_\theta). \tag{5}$$

The training loss becomes:

$$\mathcal{L}_{\text{ELBO}} = \mathcal{L}_{\text{CE}} + \beta \cdot \text{KL}\left(q(\mathbf{Z} \mid \mathbf{X}) \parallel p_\theta(\mathbf{Z} \mid \mathbf{X})\right). \tag{6}$$

Generation enhancements During inference, we apply temperature scaling and nucleus sampling to improve diversity:

$$p_T(y_t \mid \cdot) = \frac{p_\theta(y_t \mid \cdot)^{1/T}}{\sum_v p_\theta(v \mid \cdot)^{1/T}}, \quad \mathcal{V}_t = \left\{v : \sum p_T(u \mid \cdot) \leq p_{\text{nucl}}\right\}. \tag{7}$$

A coverage penalty is added to avoid repetition:

$$\mathcal{L}_{\text{cov}} = \lambda \sum_t \min(c_t, 1), \quad \text{where} \quad c_t = c_{t-1} + p_T(y_t \mid \cdot). \tag{8}$$

The final training objective combines all components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ELBO}} + \mathcal{L}_{\text{cov}} + \gamma \|\theta\|_2^2. \tag{9}$$

Evaluation objective We also explore reinforcement-style objectives directly optimizing ROUGE and BLEU:

$$\mathcal{L}_{\text{RL}} = -\mathbb{E}_{p_\theta}[R(\mathbf{Y})], \quad R = \alpha \cdot \text{ROUGE}_L + \delta \cdot \text{BLEU}, \tag{10}$$

Algorithm 1 Judicial syllogism summarization framework.

Require: Judgment document D (text)
Ensure: Abstractive summary S of D
1: $P_{\text{maj}} \leftarrow$ extract_major_premise(D) ▷ Extract legal rule or principle
2: $P_{\min} \leftarrow$ extract_minor_premise(D) ▷ Extract critical case facts
3: $S \leftarrow$ synthesize_summary(P_{maj}, $P_{\min}$, D) ▷ Generate conclusion and reasoning
4: $S \leftarrow$ post_process(S) ▷ Optional trimming/refinement
5: **return** S

yielding the hybrid loss:

$$\mathcal{L}_{\text{final}} = \mathcal{L}_{\text{total}} + \xi \cdot \mathcal{L}_{\text{RL}}. \tag{11}$$

3.2 Syllogism-guided summarization framework

Our framework consists of three main modules: Major Premise Extraction, Minor Premise Extraction, and Conclusion Synthesis. These can be executed sequentially as a pipeline or implicitly via an LLM's chain-of-thought. Figure 1 conceptually illustrates the process (major premise + minor premise → conclusion summary). Algorithm 1 provides an overview of the pipeline implementation.

Each step can be realized with a prompt to an LLM or via a task-specific model. In our implementation, all steps are handled by Qwen in a unified framework: we craft prompts such that Qwen performs steps 1-3 in one or multiple turns. Below, we explain each module and the prompt engineering involved.

3.3 Major premise extraction (Legal Rule Identification)

The first module identifies the major premise, i.e., the legal basis applicable to the case. In many judgment documents, the judges explicitly cite relevant laws or articles when providing reasoning. If the text is structured, one might locate sections like "Legal Basis" or look for law article references. Our LLM-based approach lets the model read the document and decide which law or principle is central.

We design a prompt for Qwen that asks it to focus on legal rules. For instance, a prompt template for this step might be:

```
Read the following judgment and
identify the primary legal rule
or article applied by the court in
reaching its decision.
```

The LLM then outputs a brief sentence or clause summarizing the law. We denote the extracted law text as P_{maj}. This might be a direct quote of a statute or a paraphrase (e.g., "According to Article 85 of the Contract Law, a party

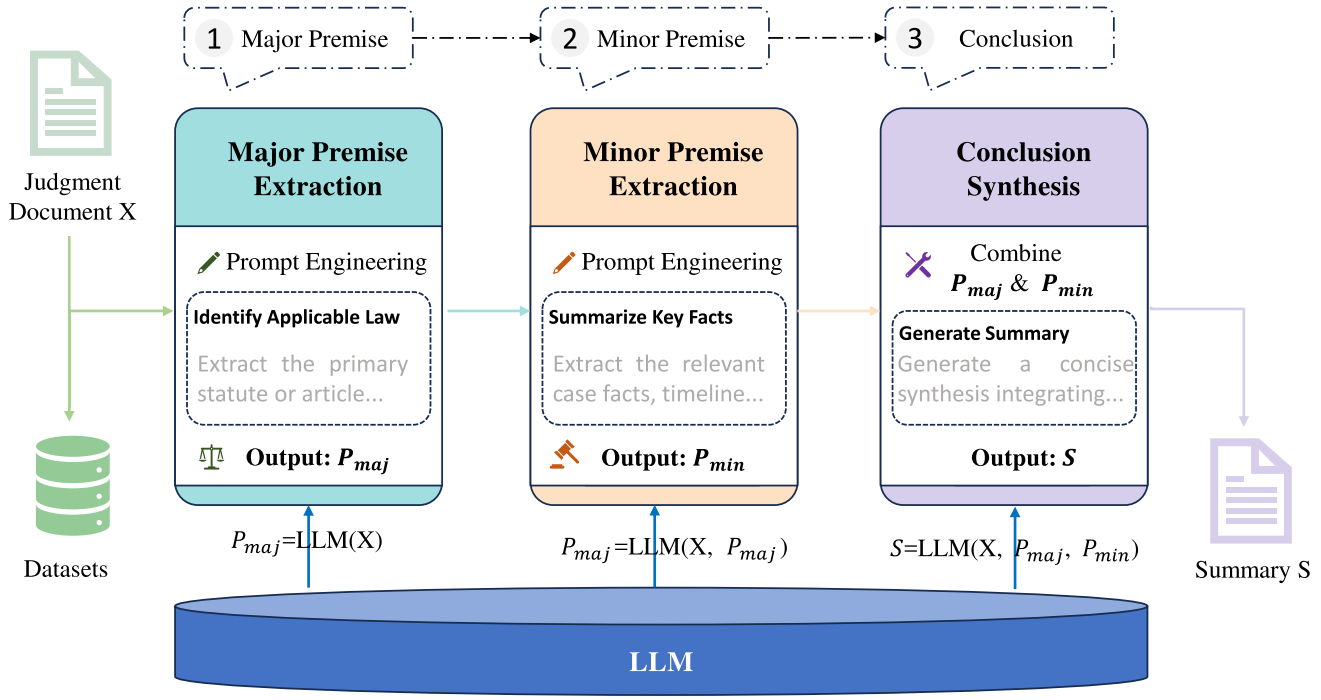


Fig. 1 Judicial syllogism summarization framework

Algorithm 2 Major premise extraction.

Require: Judgment document D
Ensure: Major premise text P_{maj} (legal rule)
1: $Q_{\text{law}} \leftarrow$ `"Identify the main law or legal rule applied: "` $+ D$
2: Provide Q_{law} to the Qwen LLM
3: Receive answer and extract it as P_{maj}
4: **return** P_{maj}

who breaches a contract must compensate the other party's losses.").

Formally, we can denote this operation as:

$$P_{\text{maj}} = \text{LLM}(I_{\text{law}}, X), \tag{12}$$

where I_{law} is an instruction prompt for law extraction, and X is the input document. P_{maj} is typically one or two sentences. In cases where multiple legal points are relevant, P_{maj} can include all, but we instruct the model to pick the most important one affecting the outcome.

In practice, step 3 may include some parsing or verification to ensure the output is indeed a legal rule statement. If the LLM produces a long answer, we may take the first sentence or the part that contains a law reference.

3.4 Minor premise extraction (Fact Summarization)

The second module distills the minor premise, i.e., the crucial facts of the case. This typically includes who did what to whom, and any conditions that trigger the law identified. Essentially, it's a summary of the case facts focusing on aspects relevant to the legal outcome. For example, in a contract dispute, the facts might include "Party A signed a contract with Party B, and Party A failed to deliver the goods by the agreed date."

We prompt Qwen to summarize the key facts. Template example:

```
Summarize the key facts of the
case that are relevant to the
court's decision.
```

The LLM will output a brief summary of the facts, which we call P_{min}. This is not the final summary, as it omits explicit mention of the law or the outcome, focusing only on factual context.

Formally:

$$P_{\text{min}} = \text{LLM}(I_{\text{fact}}, X), \tag{13}$$

where I_{fact} is the instruction for fact extraction. Optionally, the prompt might be augmented with the previously extracted P_{maj} to focus the model: e.g., "Given that the law is [P_maj], summarize the facts that are relevant." This can help the model filter out extraneous details and hone in on fact aspects that relate to the law.

Integration with chain-of-thought In an interactive chain-of-thought scenario, after obtaining P_{maj}, we include it as

context when asking for P_{min}. This mimics a judge thinking: "Knowing the law says X, what facts do we have that satisfy X?" By doing so, we reinforce the connection between the major and minor premises.

3.5 Conclusion synthesis (Summary Generation)

Finally, the conclusion module generates the summary of the judgment, incorporating the law and facts derived above and stating the outcome. This summary should logically follow from P_{maj} and P_{min}. For instance, it might produce: "*Because Party A violated the contract (failing to deliver goods), and according to Article 85 of Contract Law (major premise), the court ruled that Party A must compensate Party B (conclusion).*" This final summary S thus contains a condensed reasoning: the law, the breach of conditions, and the decision.

To produce S, we prompt Qwen with both P_{maj} and P_{min} as context (and possibly the original X for completeness), asking it to draft the judgment summary. Template:

```
Based on the following law and
facts, write a concise summary of
the judgment outcome.
Law: [P_maj]
Facts: [P_min]
Summary:*
```

Providing the structured context (law and facts) guides the LLM to integrate them. We found that including the words "Law:" and "Facts:" in the prompt followed by the content ensures the model distinctly considers both. The model then produces the "Summary:" part.

We formalize the final generation as:

$$S = \text{LLM}(I_{\text{summ}}, X, P_{\text{maj}}, P_{\text{min}}), \tag{14}$$

where I_{summ} is the instruction to compose the summary given the premises. If using a single-pass chain-of-thought approach, this final step is essentially the last part of the model's generated content after it has listed the premises internally.

Chain-of-thought vs pipeline It is worth noting that the above three modules can be executed in a single prompt by instructing the model to output each part step by step. For example, a one-shot chain-of-thought prompt might be:

```
First, state the main legal
rule. Second, state the key
facts. Finally, give the judgment
summary.
```

Algorithm 3 OneTurnCoTSummarization.

Require: Judgment document D
Ensure: Abstractive summary S of D
1: $Q_{\text{all}} \leftarrow$ `"You are a legal assistant. Read the case and summarize it step-by-step. Major Premise: (the main law applied) Minor Premise: (the key facts of the case) Conclusion: (the judgment outcome and reasoning)*" +` D
2: Provide Q_{all} to the Qwen LLM
3: Receive the output text and parse it into P_{maj}, P_{min}, and S
4: **return** S

The model's output would then be in three parts (which we could label or parse). This one-turn CoT approach ensures the model internally follows the reasoning structure. In our implementation, we experimented with both the stepwise pipeline (multiple LLM calls) and the one-turn CoT prompt. The one-turn approach (with a carefully crafted prompt) is succinct and avoids potential error accumulation across separate calls, but the multi-call pipeline allows for verifying or editing intermediate outputs. Ultimately, we used the one-turn chain-of-thought prompting for the final model for efficiency, after validating that the intermediate reasoning it produces is accurate most of the time.

In practice, we found Qwen can follow this format well, especially if we include a few-shot example of a formatted syllogistic summary in the prompt. For instance, showing one example of a case with a labeled "Major Premise / Minor Premise / Conclusion" in the prompt helps the model output our desired structure.

3.6 Implementation and training

We fine-tuned the Qwen LLM on the training portions of the datasets to further improve its summarization capability. The fine-tuning data consists of input documents with their reference summaries. To incorporate the syllogism structure during training, we experimented with two strategies:

- **Direct Fine-Tuning on Summaries:** We fine-tune Qwen on the documents paired with their normal (human-written) summaries. The model learns to map $X \rightarrow Y$ directly. During inference, we still use the chain-of-thought prompting to have it generate P_{maj}, P_{min} on the fly before Y. Even though the model wasn't explicitly trained to output the premises, we found that the prompting is sufficient for it to adapt its generation because the knowledge of how to summarize is now internalized.
- **Syllogism-augmented Fine-Tuning:** We also created an augmented training set where each target summary is prepended with an explicit breakdown. For training, we formatted each example as: "Major: [law]; Minor: [facts];

Summary: [reference summary]". We derived the [law] and [facts] for each case semi-automatically from the reference (e.g., by heuristics such as extracting any law article mentions for major, and the rest of the summary as facts and conclusion). This is noisy but provides the model an idea of the structure. Fine-tuning Qwen on this formatted data teaches it to produce syllogism-style outputs in one go. This approach, however, depends on the quality of the automatically generated premises and we used it only for a subset of data where identification was high-confidence.

We used the Adam optimizer with an initial learning rate of 2×10^{-5}, applying a linear decay schedule with 500 warm-up steps. Fine-tuning was performed for 3 epochs, totaling approximately 24,000 training steps. We set the per-device batch size to 32 and used gradient accumulation with 8 steps, resulting in an effective batch size of 256. The model with the best validation ROUGE-L was selected for final evaluation. Training was conducted on 8 NVIDIA A100 GPUs (each with 80 GB memory), using mixed-precision training and gradient checkpointing to accommodate long input sequences. The average input document length was approximately 3,000 tokens, and we used Qwen-14B with support for up to 8,000 tokens via rotary positional encoding. The total wall-clock training time was approximately 21 hours. During inference, we applied nucleus sampling with $p = 0.9$ to encourage diversity and set a maximum summary length of 200 tokens.

To summarize, the methodology integrates a logical reasoning template into the summarization process. By structuring the model's thinking into premises and conclusion, we aim to ensure that the generated summaries are not only fluent but also grounded in the legal rationale of the source text.

4 Experiments

We conduct extensive experiments to evaluate the effectiveness of our judicial syllogism summarization method. We describe the datasets used (Table 1), baseline models for comparison, evaluation metrics, implementation details, and then present the results of both the main evaluation and ablation studies. We also provide analysis and discussion of the findings.

4.1 Datasets

CAIL-SFZY dataset (Wang et al. 2023) The first dataset is the CAIL2020-SFZY judicial summary dataset, which was released as part of a competition in 2020-SFZY stands for judicial summary. This dataset contains Chinese court judgment documents paired with their official summaries. The cases cover various law domains (criminal, civil, etc.), and each summary concisely describes the case type, key facts, reasoning, and verdict. We follow the official splits: 8,000 cases for training, 1,000 for validation, and 1,000 for testing. The average judgment length is 4,000 Chinese characters, and average summary length is 300 characters. The summaries often explicitly mention law articles and the outcome, making it an ideal testbed for our syllogism approach.

Custom judgment dataset We constructed a second dataset of judicial documents to test generalization. This dataset, termed JD-CS (Judgment Documents - Custom Set), contains 5000 new cases from 2024 that are not in CAIL. These include a mix of civil contract disputes and criminal cases from public court records. Each case is annotated with a summary by law students or attorneys, following a similar style to the official ones. We use 4000 for testing and 1000 for validation (we did not fine-tune on this set to truly test generalization). The average lengths are similar to CAIL-SFZY. This dataset allows evaluating our model's performance on more recent and varied cases.

To provide more transparency regarding the legal scope of our datasets, we include detailed statistics in Table 1. The table summarizes the categories of legal cases covered, the number of distinct statutes referenced, and the sample counts for training, validation, and testing splits. These statistics support our claim that our model is evaluated across diverse legal domains.

4.2 Baseline models

We compare our Qwen-based syllogism model with several strong baselines:

- **T5** (Raffel et al. 2020): A versatile text-to-text Transformer framework trained across a wide range of NLP

Table 1 Dataset statistics

Dataset	Categories Covered	Statutes Referenced	Sample Splits
CAIL2020-SFZY	Civil (Contract, Tort)	452	7878 / 985 / 985
Custom Judgment Dataset	Criminal	1260	5000 / 800 / 800

tasks. Its task-agnostic pretraining approach enables strong performance in transfer learning scenarios.

- **BART** (Lewis et al. 2020): Integrates a bidirectional encoder with an autoregressive decoder, leveraging denoising-based pretraining. It has demonstrated strong capabilities in various text generation applications.
- **PEGASUS** (Zhang et al. 2020): Tailored for abstractive summarization, this model uses a gap-sentence generation objective during pretraining, which enhances its ability to identify and reproduce the most salient content.
- **Llama(8B)** (Touvron et al. 2023): Developed by Meta, Llama 8B is a compact yet powerful model with 8 billion parameters. Despite its relatively smaller size compared to some industry giants, Llama 8B demonstrates impressive performance on various language tasks. Its design focuses on efficiency and versatility, making it particularly suitable for environments where computational resources are limited without compromising on quality.
- **Baichuan(7B)** (Yang et al. 2023): Available in both 7 billion and 14 billion parameter versions, are tailored to meet the demands of multilingual and domain-specific applications. These models emphasize a balance between speed and accuracy, providing strong performance in generating and understanding text. Their dual-variant approach offers users the flexibility to choose a model that best fits their computational and task-specific requirements.
- **Yi(9B)** (Young et al. 2024): The Yi 9B model is built with 9 billion parameters and is designed to deliver high-quality performance across a spectrum of natural language processing tasks. It focuses on producing coherent and contextually relevant outputs, making it a competitive choice for applications that demand both precision and fluency in language generation.
- **Gemma(7B)** (Team et al. 2024): Gemma offers two variants-one with 7 billion parameters and another with 12 billion-each optimized for different operational scenarios. These models are designed to provide a balance between speed and capability, with the flexibility to scale according to task complexity. Gemma's architecture is particularly effective in generating concise and accurate summaries, as well as handling other generative tasks.
- **Phi(7B)** (Abouelenin et al. 2025): The Phi models come in two sizes, 7 billion and 14 billion parameters, offering users options based on their performance and resource requirements. Phi is engineered to excel in both understanding and generating natural language, with its larger variant delivering enhanced depth in language comprehension. This versatility makes it well-suited for a broad array of applications in natural language processing.

4.3 Evaluation metrics

We evaluate summarization quality using several standard metrics:

- **ROUGE** (Lin 2004) **(Recall-Oriented Understudy for Gisting Evaluation):** We report ROUGE-1, ROUGE-2, and ROUGE-L (F_1 scores) which respectively measure the overlap of unigrams, bigrams, and longest common subsequence between the system summary and reference summary. ROUGE is a recall-heavy metric commonly used in summarization. Specifically, ROUGE-N precision and recall are defined as:

$$\text{Precision}_{R,N} = \frac{\#\{matching N-grams\}}{\#\{N-gramsinsystemsummary\}} \tag{15}$$

$$\text{Recall}_{R,N} = \frac{\#\{\text{matching N-grams}\}}{\#\{\text{N-grams in reference summary}\}}, \tag{16}$$

and the F_1 combines them. ROUGE-L measures overlap of longest common subsequences, capturing fluency and in-order coverage.
- **BLEU** (Papineni et al. 2002) **(Bilingual Evaluation Understudy):** Although BLEU is primarily used in machine translation, we include it to gauge overlap with reference in a precision-oriented manner. We compute BLEU-4 (up to 4-gram) with the standard formulation involving modified n-gram precision p_n and a brevity penalty (BP) for shorter summaries. In formula form:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{4} w_n \ln p_n\right), \tag{17}$$

where we use uniform weights $w_n = 1/4$. $\text{BP} = \min(1, e^{1-|Y^|/|Y|})$ penalizes too-short outputs ($|Y|$ = length of system summary, $|Y^|$ = length of reference).
- **BERTScore** (Zhang et al. 2019): This metric uses pretrained contextual embeddings to measure semantic similarity between summary and reference. We use BERTScore F_1, which computes token-wise similarity and is robust to paraphrasing. Given the set of tokens in system summary Y and reference Y^*, with each token y represented by a contextual embedding $\mathbf{e}(y)$ (from RoBERTa or Chinese BERT), we define precision and recall as:

$$\text{Prec}_{BS}(Y, Y^*) = \frac{1}{|Y|} \sum_{y \in Y} \max_{y^* \in Y^*} \sin(\mathbf{e}(y), \mathbf{e}(y^*)), \tag{18}$$

$$\mathrm{Rec}_{BS}(Y, Y^*) = \frac{1}{|Y^*|} \sum_{y^* \in Y^*} \max_{y \in Y} \sin(\mathbf{e}(y^*), \mathbf{e}(y)), \quad (19)$$

where $\mathrm{sim}(a, b)$ is cosine similarity. The BERTScore F_1 is the harmonic mean of Prec and Rec. We used the multilingual model for embeddings due to Chinese content.

These metrics provide a multi-faceted evaluation: ROUGE emphasizes coverage of reference information (important in legal contexts to not omit facts), BLEU emphasizes precision (no extra unrelated info), and BERTScore checks semantic alignment.

4.4 Results on main evaluation

Tables 2 and 3 summarize the performance of our new model ("Our") against five competitive baselines on the CAIL and JD-CS test sets. All ROUGE and BLEU scores are reported as F1 percentages; BERTScore is the F1 similarity score.

Our model outperforms all baselines by a substantial margin. In particular, it achieves a ROUGE-L of 64.35%, which is roughly 4.3 points higher than the strongest baseline (LLaMa8B). The ROUGE-1 and ROUGE-2 improvements (≈2.6 and 3.1 points over LLaMa8B, respectively) indicate superior capture of unigrams and bigrams, reflecting both better content coverage and phrasing alignment. The BLEU score rise to 58.77% (an increase of 4.5 points) suggests our summaries are more precise with fewer unnecessary tokens. Finally, the BERTScore gain to 0.8698 confirms stronger semantic fidelity to reference summaries.

Among the baselines, LLaMa8B consistently leads the non-syllogistic group, reflecting the benefits of its parameter scale and fine-tuning. Models like Phi7B and Gemma9B lag behind, particularly on bigram-level metrics, suggesting that smaller architectures struggle to capture detailed legal phrasing even after fine-tuning.

The cross-dataset evaluation on JD-CS confirms the generalizability of our approach. Although all models exhibit a modest drop compared to their CAIL scores-likely due to domain shift-the ranking remains the same. Our method again secures the top spot across every metric, with a ROUGE-L of 63.98%, outperforming LLaMa8B by about 4.5 points. The consistent gains in ROUGE-1/2 and BLEU indicate robust content extraction and precision even on unseen case types. The BERTScore advantage (≈0.89 over baselines) underscores strong semantic alignment in a cross-domain setting.

These results demonstrate that integrating judicial syllogism prompting with domain-specific fine-tuning yields substantial improvements over both zero-shot and directly fine-tuned baselines. The structured reasoning framework not only enhances recall of key facts (higher ROUGE-1/2) but also sharpens summary precision (higher BLEU), culminating in the strongest semantic similarity (highest BERTScore). Across two distinct legal datasets, our model establishes a new state-of-the-art, underscoring the value of combining large-scale architectures with task-tailored reasoning modules.

4.5 Ablation studies

To better understand the contribution of each part of our approach, we performed ablation experiments on the validation set:

- **No Major Premise Prompt:** We removed the explicit law prompting. The model was instructed to summarize the case focusing only on facts and outcome. This tests whether explicitly including the legal premise is helpful.
- **No Minor Premise Prompt:** We removed the explicit step focusing on facts, and only asked the model to provide a law and conclusion.
- **No Chain-of-Thought (Direct):** The model was asked to directly generate the summary without any step-by-step structure (this is effectively the Qwen-Direct baseline).
- **One-turn vs Multi-turn:** We compared the one-turn CoT prompting with a multi-turn pipeline. Both use the law+fact structure but one is single prompt and the other is multi-prompt.

Table 2 Results on CAIL-SFZY dataset

Model	Rouge-1	Rouge-2	Rouge-L	BLEU	BERTScore
T5	56.24	35.88	53.25	45.45	72.45
BART	49.36	21.55	39.56	34.54	75.64
PEGASUS	50.45	22.65	40.56	38.55	78.65
LLaMa8B	62.38	43.02	60.05	54.24	86.14
Phi7B	59.01	38.72	56.55	49.82	81.70
Yi9B	59.61	39.26	58.15	50.99	83.07
Baichuan7B	60.95	41.83	59.03	54.20	84.78
Gemma9B	58.16	37.88	56.30	49.52	81.43
Our	**64.99**	**46.11**	**64.35**	**58.77**	**86.98**

Table 3 Results on custom judgment dataset

Model	Rouge-1	Rouge-2	Rouge-L	BLEU	BERTScore
T5	55.13	34.78	52.17	44.42	70.15
BART	48.16	20.57	38.24	33.12	74.13
PEGASUS	49.05	21.63	39.57	37.75	76.15
LLaMa8B	61.74	42.49	59.50	53.55	85.44
Phi7B	58.33	38.08	56.32	49.54	81.21
Yi9B	58.71	38.71	57.38	50.86	82.16
Baichuan7B	60.58	41.51	58.85	53.30	84.30
Gemma9B	57.92	37.64	56.04	48.86	80.47
Our	**64.73**	**45.36**	**63.98**	**58.39**	**86.33**

Tables 4 and 5 report the performance of our full syllogism-based summarization approach and its variants on the CAIL and JD-CS datasets, respectively. Across both benchmarks, removing any component leads to a clear degradation in automatic metrics, confirming that each part of our design contributes positively to overall performance.

Our ablation study on both the CAIL and JD-CS datasets demonstrates that every component of our syllogism-based prompting framework contributes significantly to summarization quality. When we removed the explicit law premise, performance plunged-ROUGE-1 scores fell by nearly 8 points on CAIL (from 64.99 to 57.10) and by a similar margin on JD-CS, while BLEU dropped by over 7 points in both cases. This sharp decline highlights that providing the statutory foundation is essential for grounding the model's focus in the correct legal framework. Equally important is the fact-premise step. Omitting the prompt that extracts case facts leads to comparable degradations: on CAIL, ROUGE-1 drops from 64.99 to 56.13, and BLEU from 58.77 to 50.76; on JD-CS, ROUGE-1 falls to 55.91 and BLEU to 50.43. Without structuring the extraction of factual details, the model struggles to weave them coherently into its legal reasoning, thereby undermining summary accuracy. The chain-of-thought (CoT) scaffold further amplifies these gains. The direct-summary baseline (no CoT) yields ROUGE-1 scores around 55-56 and BLEU scores around 50 on both datasets-roughly 9 points below our full model. This indicates that step-by-step reasoning not only ensures more complete integration of premises but also enhances fluency and consistency in the final summary. Comparing one-turn and multi-turn implementations of CoT reveals a practical trade-off. The multi-turn pipeline recovers much of the lost performance (achieving ROUGE-1 $\approx$ 59.5), yet the single-prompt, one-turn CoT matches or slightly exceeds it (ROUGE-1 $\approx$ 60.1) without the overhead of iterative interactions. This suggests that embedding the entire reasoning chain in a single prompt is both efficient and effective.

In sum, our results validate the key design choices of our approach: explicit law and fact premises are indispensable, and a structured CoT framework markedly boosts summarization performance. Moreover, one-turn CoT offers a streamlined alternative to more complex multi-turn pipelines, striking an optimal balance for practical deployment.

4.6 Low-resource performance analysis

To further evaluate the robustness and adaptability of our proposed judicial syllogism framework, we conducted a series of low-resource experiments using only 50, 100, and 200 training examples. This setting simulates practical deployment scenarios where large-scale legal annotations may not be available. We compare the performance of our method with the base summarization model under these constraints. The results are reported in Tables 6 and 7 for the CAIL-SFZY and Custom Judgment datasets, respectively.

Table 4 Ablation studies results on CAIL-SFZY dataset

Model	Rouge-1	Rouge-2	Rouge-L	BLEU	BERTScore
Full Syllogism (Ours)	64.99	46.11	64.35	58.77	86.98
- no Law premise	57.10	40.51	56.54	51.64	85.93
- no Fact premise	56.13	39.82	55.58	50.76	85.71
- no CoT (direct summary)	55.64	39.48	55.09	50.31	85.46
Multi-turn pipeline	59.48	42.20	58.89	53.79	86.30
One-turn CoT (single prompt)	60.14	42.67	59.55	54.38	86.45

Table 5 Ablation studies results on custom judgment dataset

Model	Rouge-1	Rouge-2	Rouge-L	BLEU	BERTScore
Full Syllogism (Ours)	64.73	45.36	63.98	58.39	86.33
- no Law premise	56.87	39.85	56.21	51.30	85.29
- no Fact premise	55.91	39.18	55.26	50.43	85.07
- no CoT (direct summary)	55.42	38.83	54.78	49.99	84.82
Multi-turn pipeline	59.24	41.51	58.56	53.44	85.60
One-turn CoT (single prompt)	59.90	41.97	59.21	54.03	85.70

As shown in Table 6, our model consistently outperforms the base method across all resource levels. When trained on only 50 examples, our approach achieves ROUGE-1/2/L scores of 55.15/42.14/44.14, outperforming the base by approximately 2 points on all metrics. As the number of training examples increases to 200, our method reaches 59.81 (R1), 45.65 (R2), and 47.98 (RL), maintaining a notable lead over the base model (57.01/43.57/44.96). These results indicate that the structured reasoning embedded in our approach facilitates better generalization, even in data-scarce settings.

Table 7 presents similar trends on the Custom Judgment Dataset. While all models show lower performance due to domain shift and lack of fine-tuning, our method demonstrates consistent superiority. For example, with only 200 examples, our model achieves ROUGE-1/2/L scores of 54.65/44.10/43.45, exceeding the base model by 2.5-3 points. These results reinforce the robustness of the judicial syllogism framework across datasets and legal domains.

4.7 Further analysis and discussion

Qualitative analysis We examined sample outputs to understand how the syllogism approach affects the content. In one case about a fraud crime, the reference summary mentioned the specific law violated (Article XYZ of Criminal Law) and how the defendant's actions satisfied the conditions of that crime, leading to the sentence. Our model's summary similarly included that law article and a concise explanation of the acts, whereas LLaMa8B's summary generically stated the defendant committed fraud but did not cite the law. This likely explains the higher ROUGE (the reference also cites the law). In another civil case about a loan dispute, the syllogism model explicitly noted the contract clause and that the defendant failed to repay, leading the court to order repayment with interest - again matching reference details. The direct baseline, in contrast, gave a shorter summary that the defendant must repay the loan, but omitted mention of interest and the clause, missing some points. These examples illustrate that guiding the model to include both the legal basis and facts yields more thorough summaries.

Error analysis We did notice some failure modes. Occasionally, the model might infer a wrong law if the judgment text was implicit. For example, if the judgment didn't explicitly cite a statute, the model sometimes guessed (occasionally incorrectly) which law applied. This is rare and usually the model chooses a reasonable related law, but it could introduce inaccuracies. One possible remedy is to integrate a legal retrieval step: e.g., query an external database for relevant laws given the case facts (which could be an extension of our pipeline). Another issue is verbosity: sometimes the chain-of-thought prompt made the model output a bit lengthier summary including both premises and conclusion explicitly. If brevity is desired, a tuning of prompt or a post-process to remove redundant phrases is needed. We tuned our prompt to emphasize "concise summary" to handle this.

Generality of approach While our experiments focus on Chinese judgments, the approach is language-agnostic and domain-agnostic in principle. The idea of breaking a sum-

Table 6 Experimental results under low-resource summarization on CAIL-SFZY dataset

Examples	Method	Rouge-1	Rouge-2	Rouge-L
50	Base	53.28	40.14	42.15
	Our	55.15	42.14	44.14
100	Base	56.94	43.17	44.25
	Our	59.47	45.15	47.64
200	Base	57.01	43.57	44.96
	Our	59.81	45.65	47.98

Table 7 Experimental results under low-resource summarization on custom judgment dataset

Examples	Method	Rouge-1	Rouge-2	Rouge-L
50	Base	50.14	38.45	39.04
	Our	51.74	40.35	40.85
100	Base	51.25	39.25	40.15
	Our	52.36	42.14	41.24
200	Base	52.11	40.51	41.04
	Our	54.65	44.1	43.45

mary task into premise and conclusion can apply to other domains that have an underlying logical structure. For example, summarizing a scientific paper might involve identifying hypothesis, method, result (analogous to premises and conclusion). Our results suggest that injecting reasoning structure helps the model focus on why something is the outcome, which could be beneficial in any complex summarization task. In the legal domain, an interesting future direction is to incorporate multiple minor premises (for multi-faceted cases) or to handle scenarios where the major premise itself may be a composite of multiple laws. The current framework handles the primary law; extending it to multi-law cases could involve iterative application of the syllogism if needed.

5 Conclusion

We presented a novel approach to legal text summarization that leverages judicial syllogism with large language models. By structuring the summarization process into identifying the relevant law (major premise), extracting key facts (minor premise), and then generating the conclusion, our method guides LLMs to produce summaries that are rich in legal reasoning and factual details. We implemented this approach with the Qwen 14B model, using chain-of-thought prompting to emulate a judge's logical reasoning within the model's generation process. Our experiments on Chinese judgment datasets demonstrate that this syllogism-driven method yields state-of-the-art results, significantly outperforming powerful baselines. The improvements in ROUGE, BLEU, and BERTScore confirm that our summaries more closely match the ground truth in content and semantics. Through ablation studies, we validated that each component of the syllogistic reasoning contributes to the performance, and that the approach is robust. The success of our model highlights the benefit of injecting domain-specific reasoning structures into LLM-driven text generation. In conclusion, the integration of legal reasoning patterns (like syllogism) with advanced LLMs opens up a promising pathway for explainable and accurate summarization. The summaries generated are not just accurate but also interpretable, as they inherently explain the why behind the outcome. This approach can enhance the transparency of AI in legal applications, which is valuable for user trust.

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Data Availability Statement The datasets used in this manuscript are publicly available datasets. Detailed information about these datasets is provided in Section Dataset of this manuscript.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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